



POLITECNICO
MILANO 1863

FPGA-Based Lane Detection and Closed-Loop Lane Keeping Control for Autonomous Vehicles

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Summary

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Introduction

01

Project scope & Validity assumptions

Il progetto ha l'obiettivo di sviluppare un sistema di lane detection e lane keeping in anello chiuso, con successiva valutazione dell'impatto dell'introduzione di un'architettura FPGA per Image Processing e Curve Fitting, valutando pro e contro nell'adozione di diversi approcci ed algoritmi.

- **Telecamera frontale montata rigidamente** sul veicolo: posizione ed orientamento rispetto al sistema di riferimento del veicolo non variano durante la simulazione;
- **Parametri intrinseci** (lunghezza focale e centro ottico), **ed estrinseci** (altezza, pitch, yaw e posizione rispetto al veicolo) sono **noti** e fissati nel modello;
- **Velocità** del veicolo pressoché **costante**;
- **Sideslip trascurabile**, per la validità del modello cinematico, in cui inoltre la dinamica longitudinale e laterale sono disaccoppiate;
- Il **raggio di curvatura** della traiettoria è **molto maggiore delle dimensioni caratteristiche del veicolo**, in particolare passo e carreggiata, così per la validità del Bicycle Model;
- In ambiente Simulink si assumono **disponibili direttamente le grandezze necessarie** al controllo (velocità longitudinale, heading/yaw rate e angolo di sterzo);
- La **larghezza della corsia** è assunta **costante** e nota;
- La validazione iniziale è effettuata in **condizioni visive controllate**, senza forti ombre, occlusioni delle corsie da parte di altri veicoli o ostacoli, pioggia, riflessi o marcatore stradali degradate, oltre che su una strada piana, senza salite o discese.

Closed-loop system on Simulink

02

Model-based Image Processing Subsystem

Components:

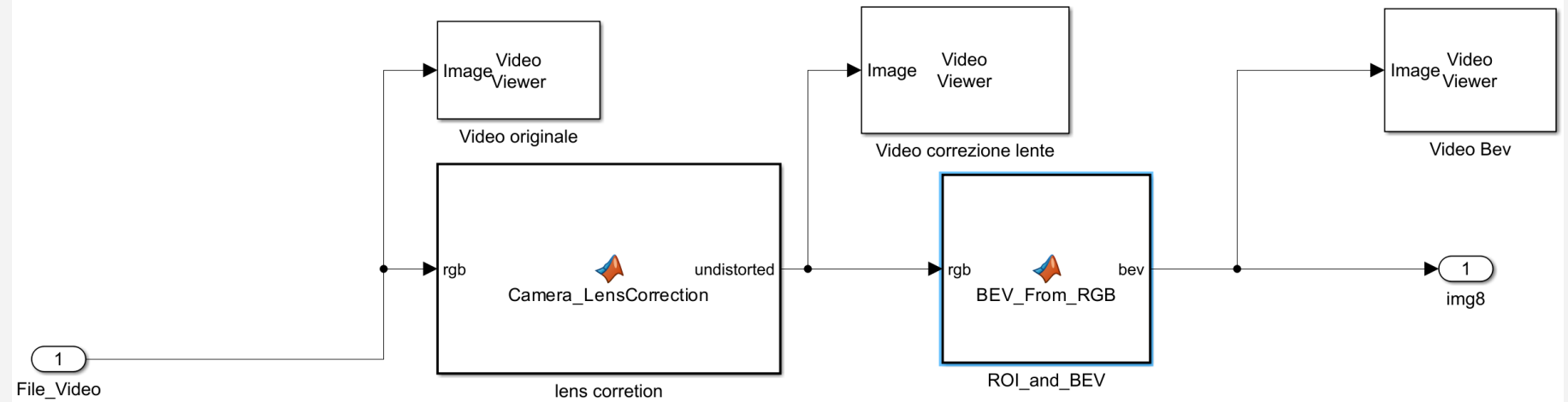
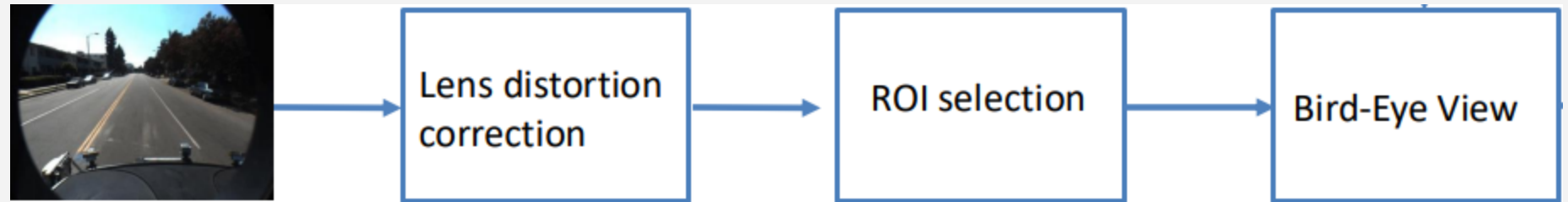
- Lens distortion correction block
- ROI and BEV block

Input:

- File Video (single frame) from Simulation 3D Camera (Automated Driving Toolbox)

Output:

- BEV frame



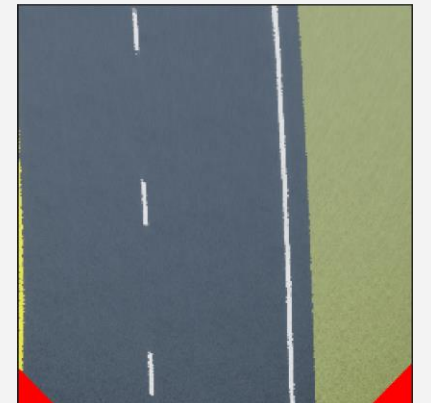
Subsystem dell'immagine processing



Initial frame



Lens distortion correction



ROI and BEV frame

Lane Segmentation

Components:

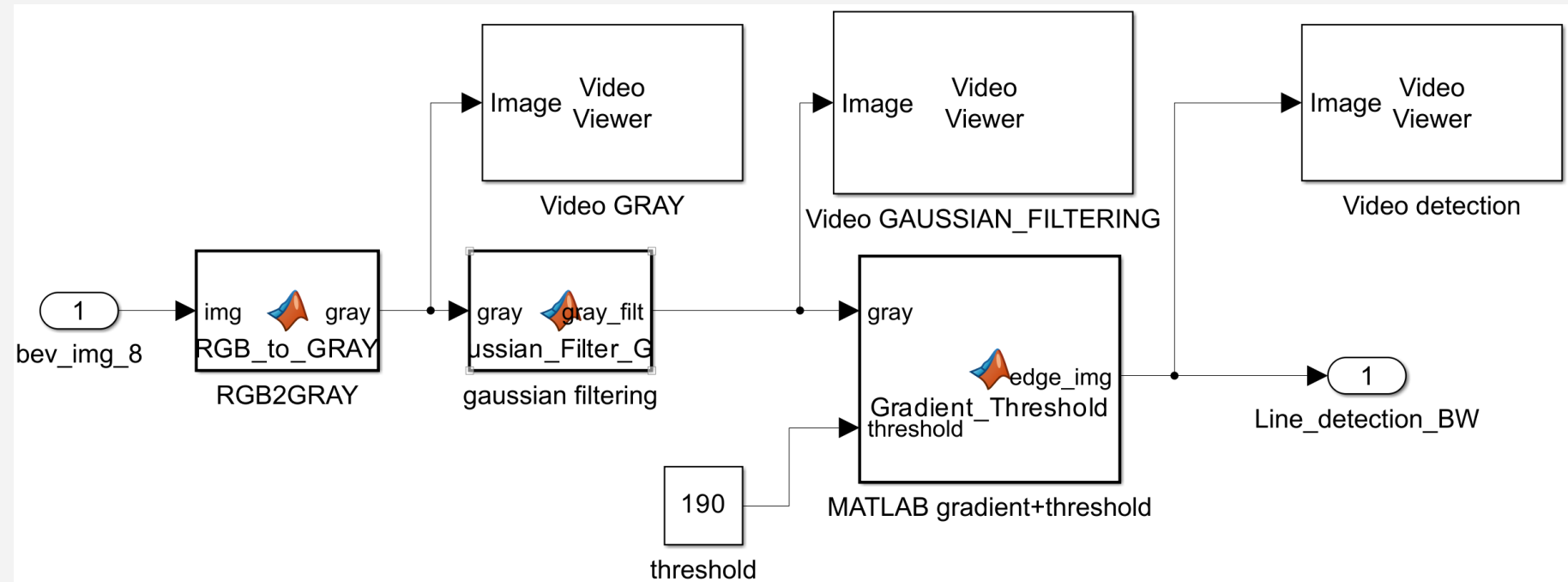
- RGB2GRAY
- Gaussian filtering
- Gradient + Thresholding

Input:

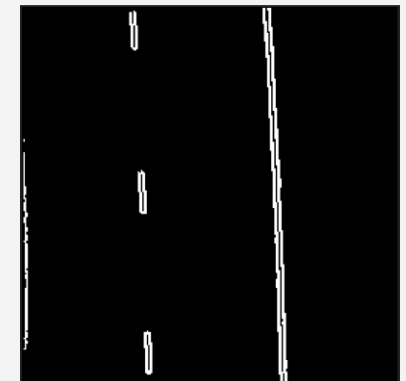
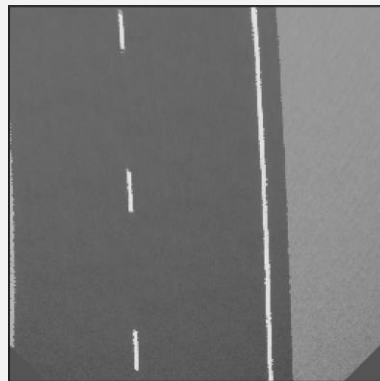
- BEV frame

Output:

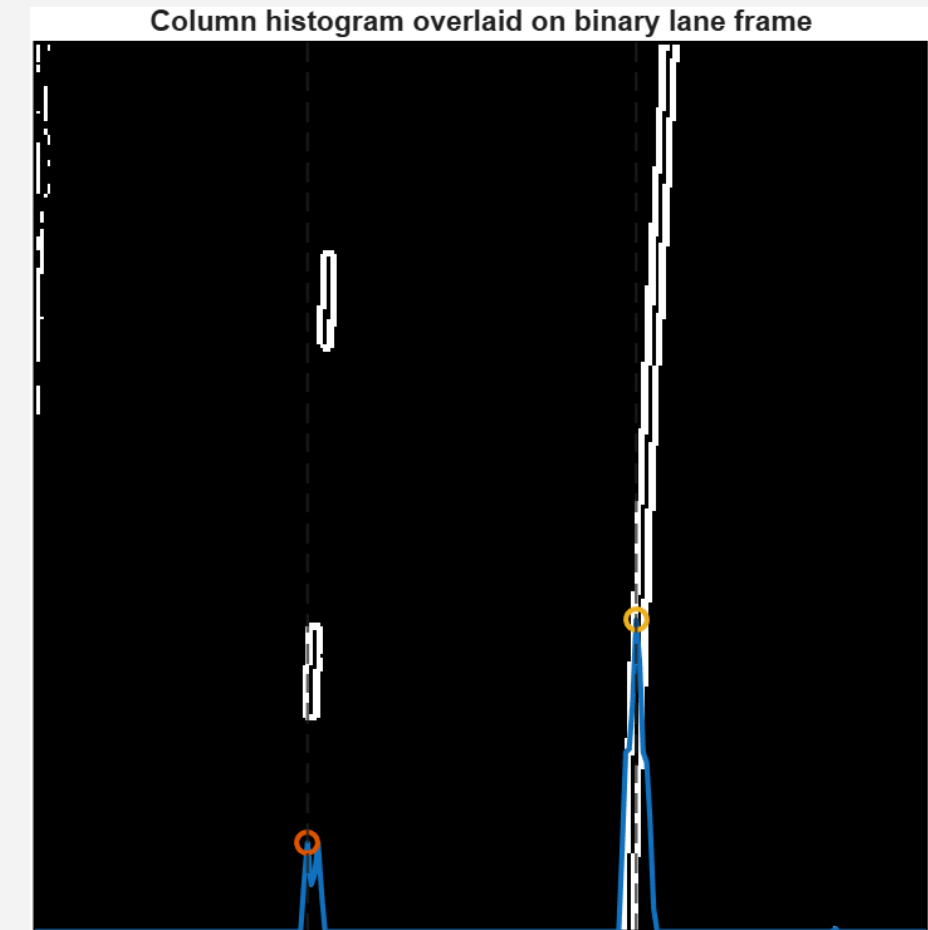
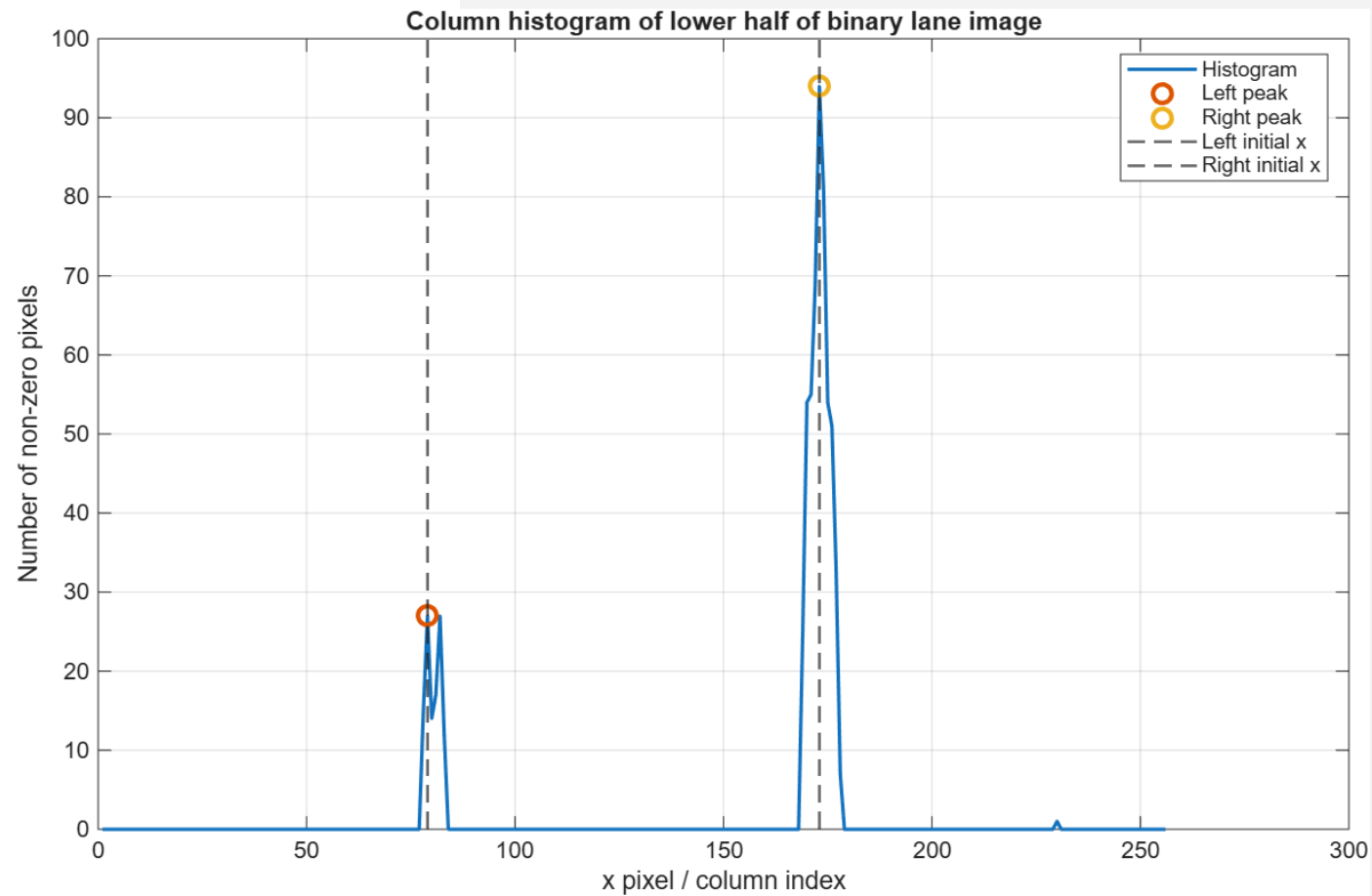
- Image ready for curve fitting



Subsystem del LINE DETECTION



Lane Segmentation: Results



Curve Fitting and Lateral Control

Components

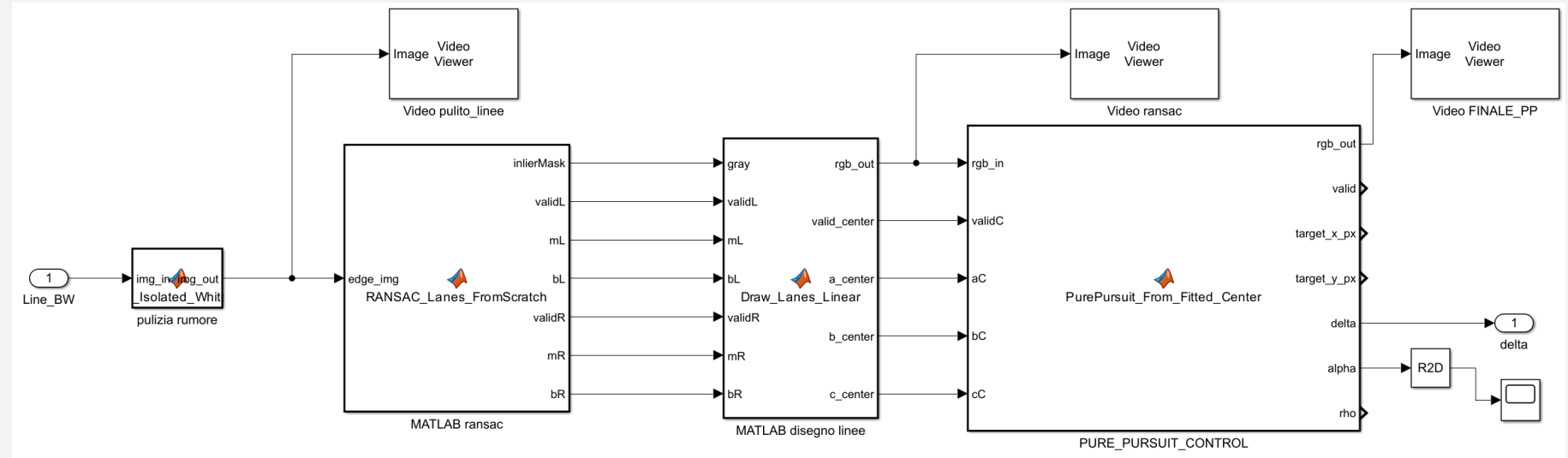
- pulizia rumore
- Lane mask corsia
- rotazione corsia
- PURE PURSUIT

Input:

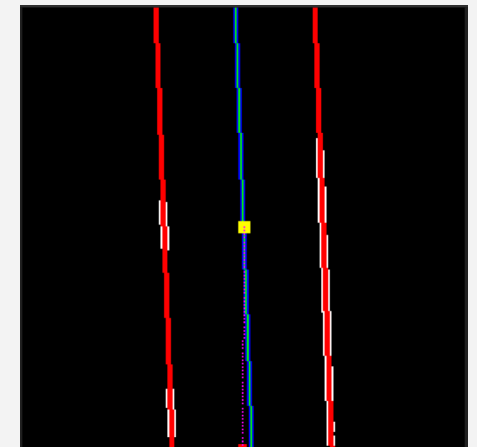
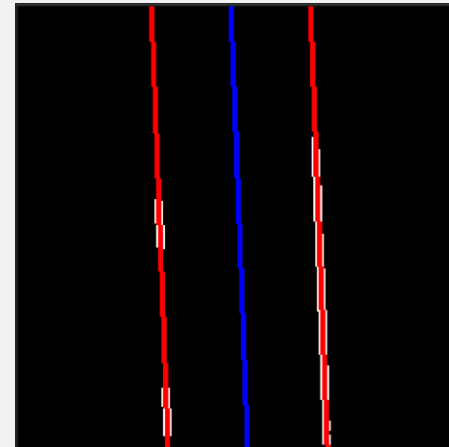
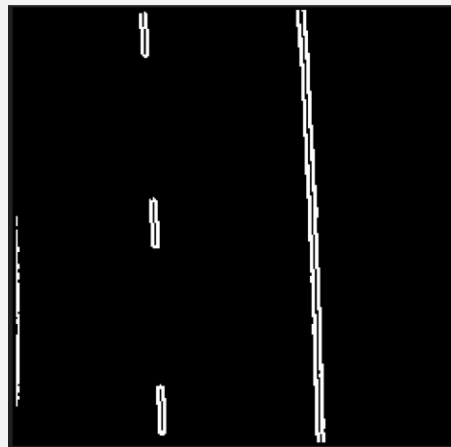
- Line BW

Output:

- delta



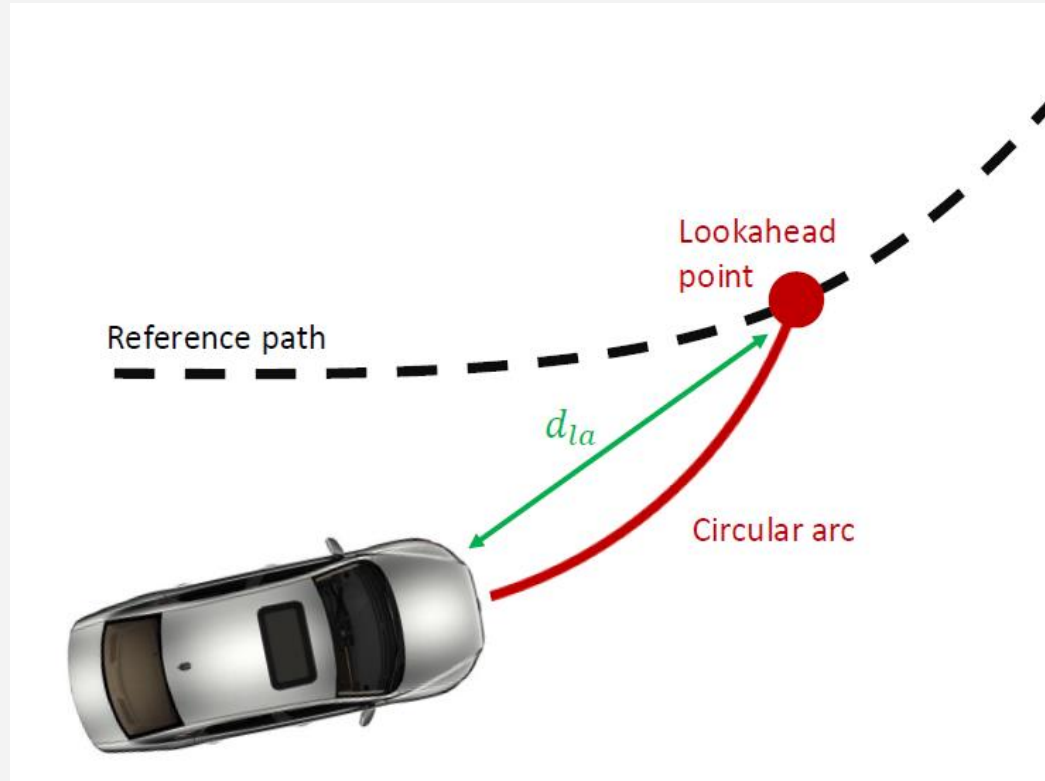
Subsystem del LANE_ KEEPING_ CONTROL



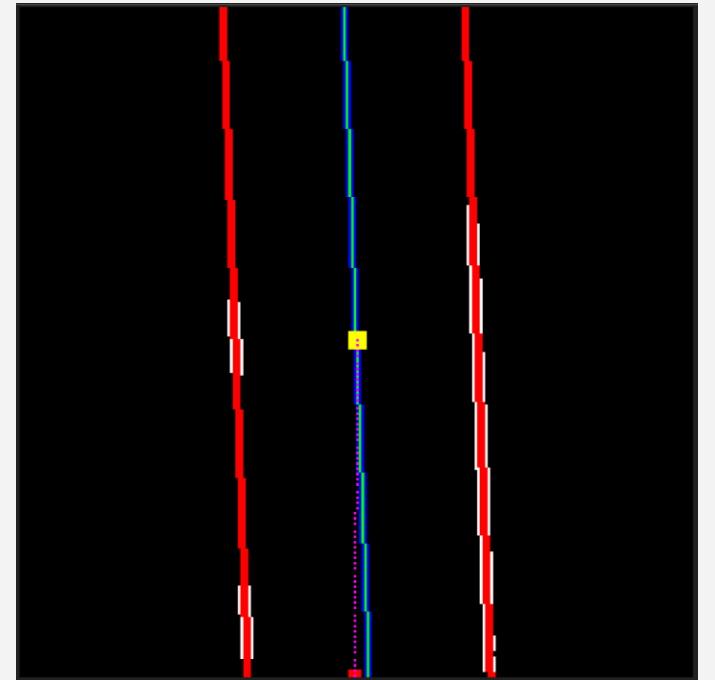
Pure Pursuit Description

Passages:

- 1) Link the vehicle position to the lookahead point using a circular arc.
- 2) Compute the geometric steering angle necessary to follow the circular path.



Pure_pursuit: **geometric** trajectory tracking using a look-ahead point



Pure_pursuit implementation on simulink

Bycycle Model

Components

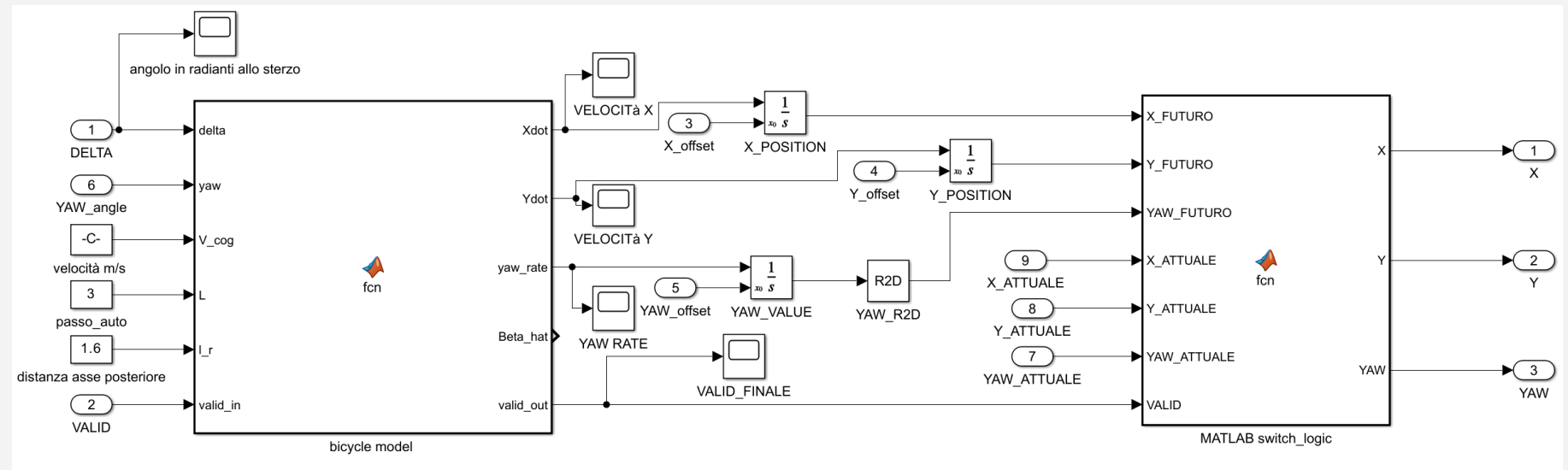
- Constants
- Bicycle model

Inputs:

- Delta
- Yaw angle
- COG Speed (136,5Km/h) 37,92 m/s

Outputs:

- X
- Y
- psi



Subsystem del Bycycle Model

Bycycle Model Equations

Kinematic Bicycle model

X, Y : global coordinates

ψ : heading

r : yaw rate

δ : steering angle

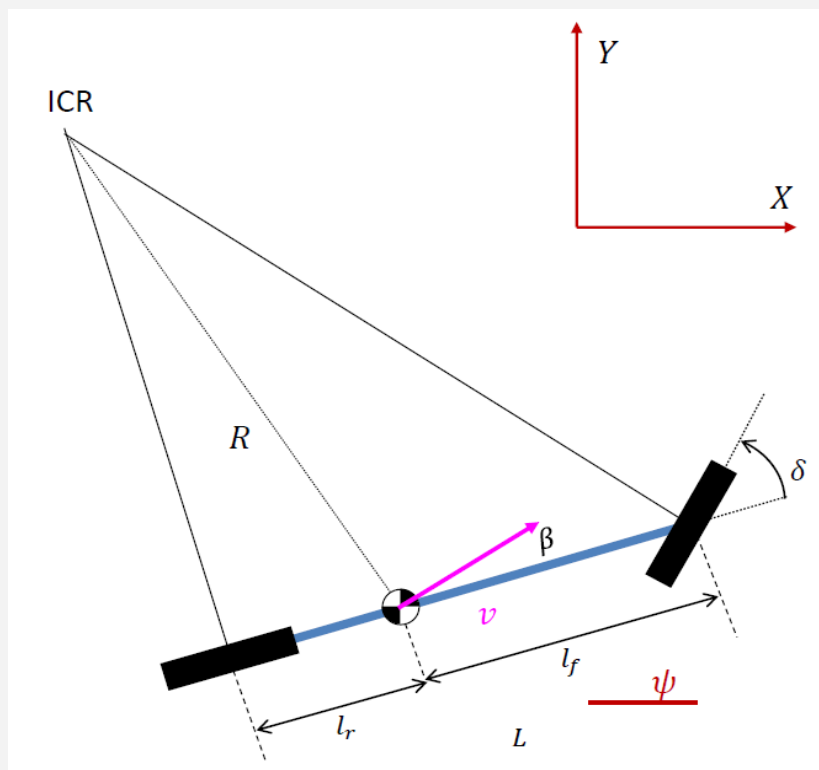
V : COG velocity $\leftarrow$ **constant**

$$\dot{X} = V \cos(\hat{\psi} + \hat{\beta})$$

$$\dot{Y} = V \sin(\hat{\psi} + \hat{\beta})$$

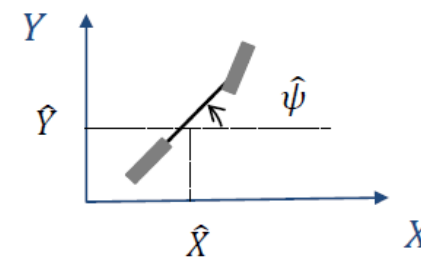
$$r = \frac{V}{L} \tan(\delta) \cos(\hat{\beta})$$

$$\hat{\beta} = \text{atan}\left(\frac{l_r}{L} \tan(\delta)\right)$$



Vehicle Bicycle models

Kinematic or dynamic bicycle model



Vehicle Bicycle model inputs and outputs



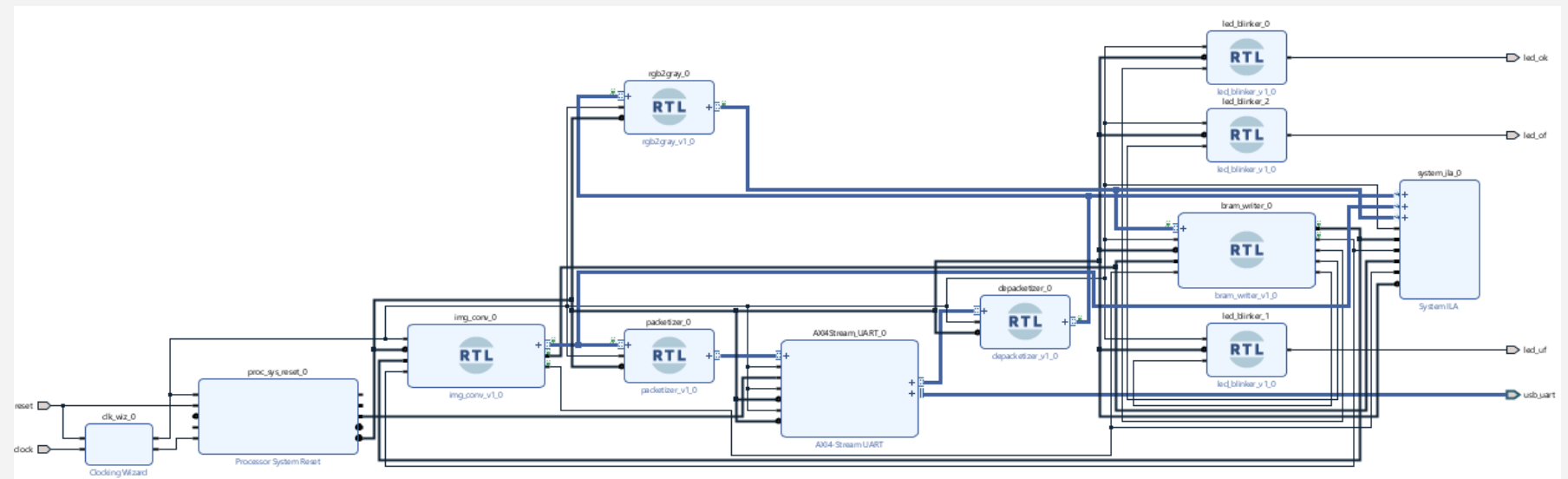
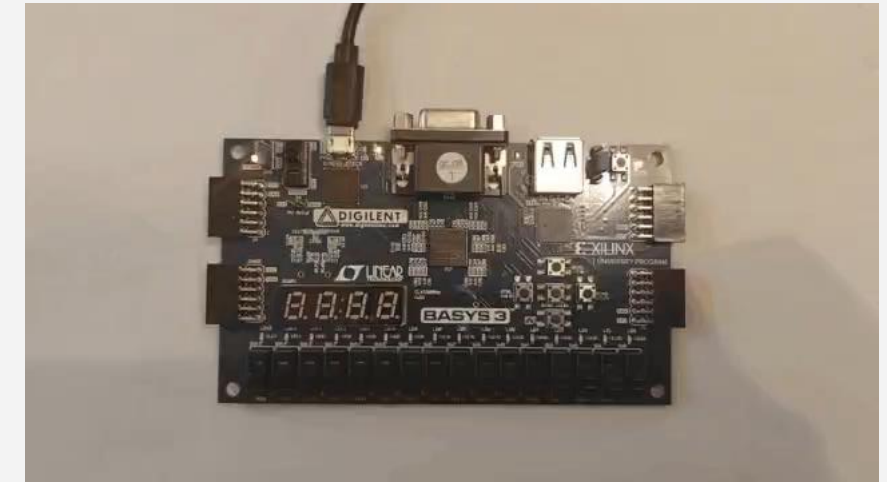
FPGA in the loop

03

FPGA for line detection



Digilent Basys 3 Artix-7 FPGA



Vivado Block design of the complete VHDL of the line detection

FPGA Simulink connection

Components

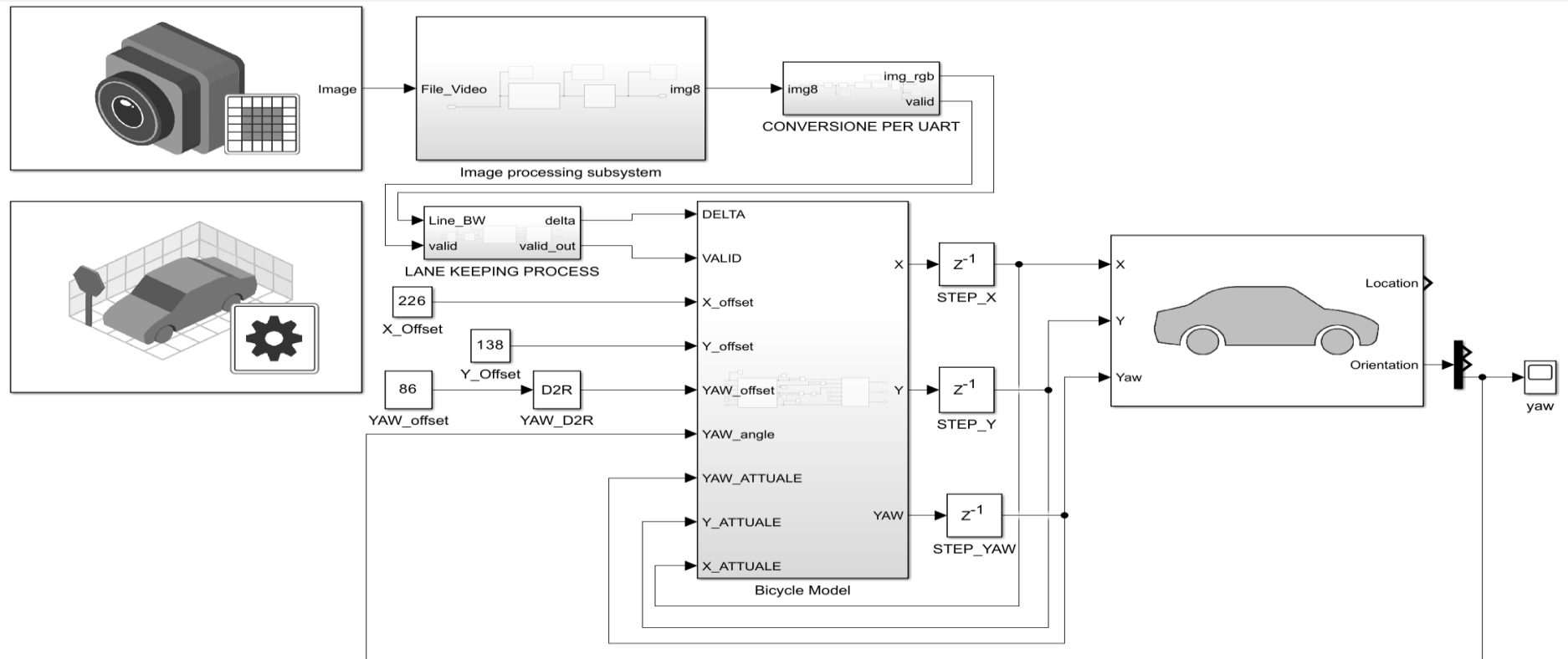
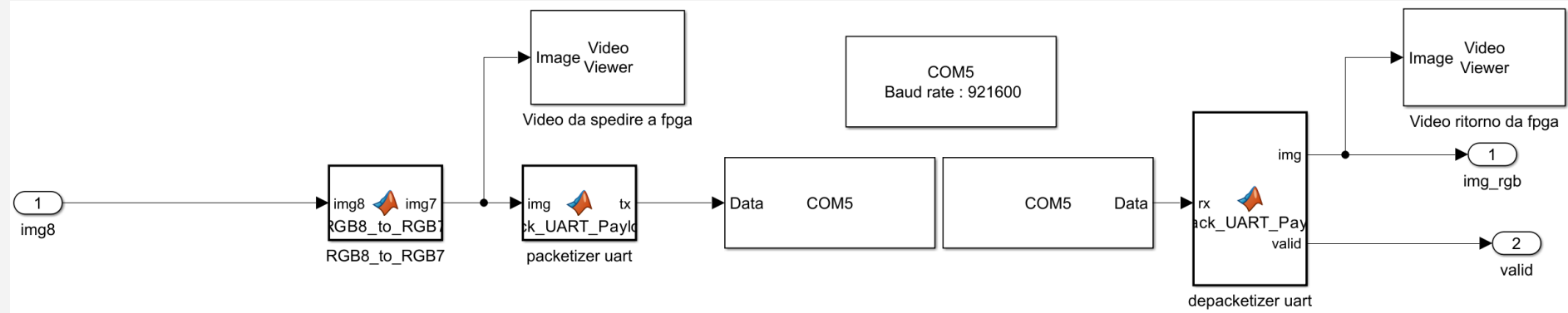
- RGB8 to RGB7
- Packetizer UART
- Depacketizer UART

Inputs:

- img8

Outputs:

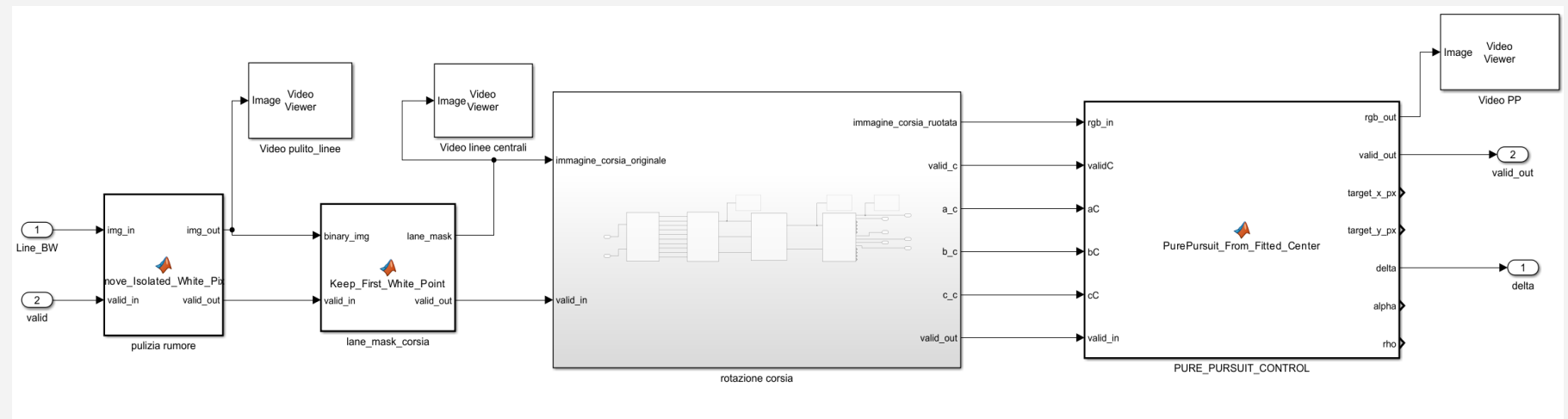
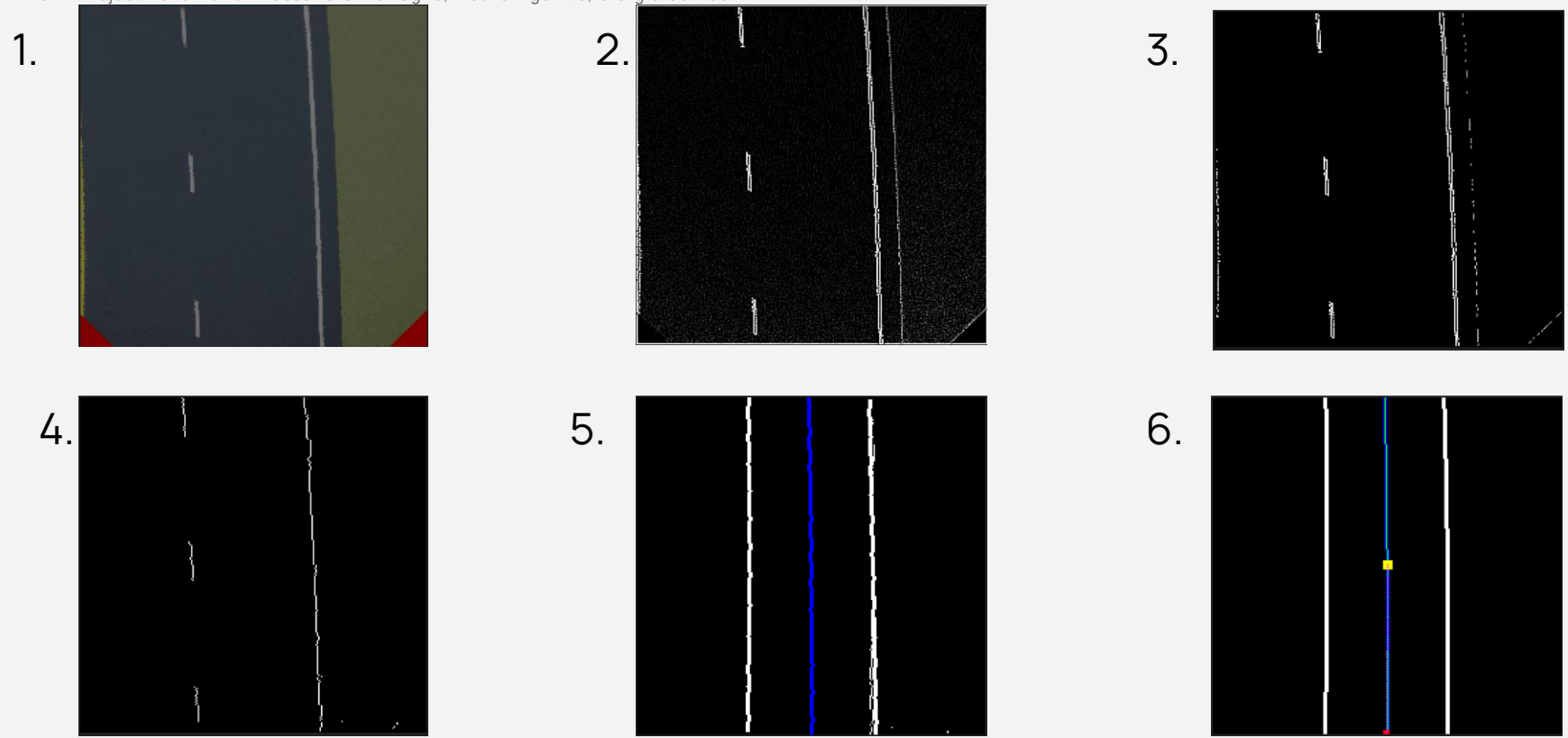
- Img gray
- valid



FPGA FRAME STEPS

Components

1. FRAME TO FPGA
2. FRAME FROM FPGA
3. FRAME FILTERED
4. FRAME LINE MASK
5. FRAME ROTATED
6. FRAME PURE PURSUITE





Conclusions

04

Simulation Straight Road

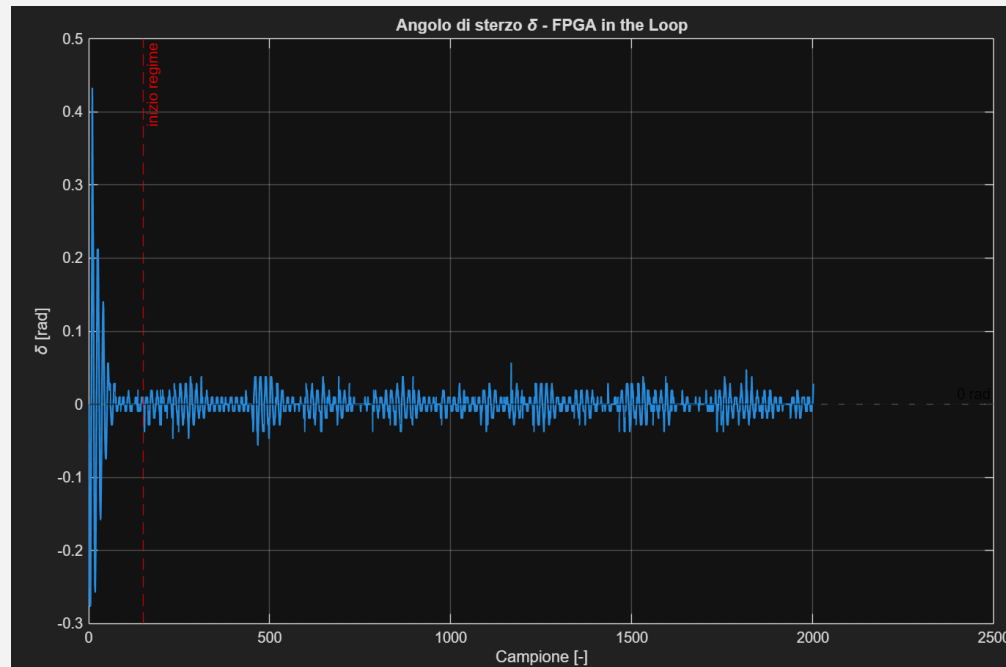
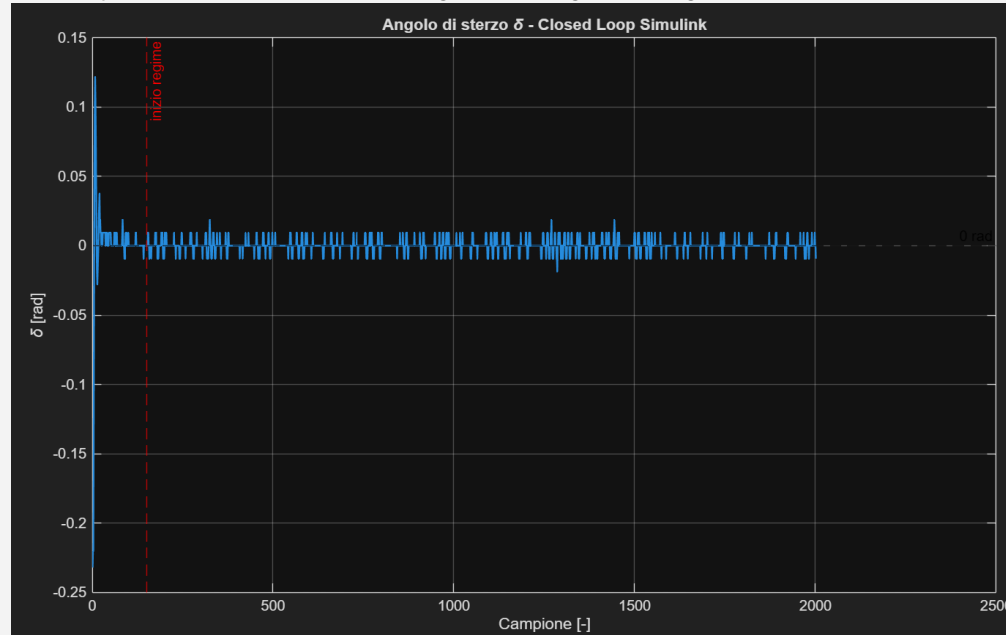


CLOSED LOOP FPGA QUADRATIC FIT



CLOSED LOOP SIMULINK RANSAC

RANSAC VS QUADRATIC MODEL



Closed Loop Simulink:

```

----- Statistiche globali: campioni 1:end -----
Numero campioni      = 2001
Media                 = 0.000005 rad
Valore assoluto medio = 0.002889 rad
RMS                   = 0.011425 rad
Massimo assoluto      = 0.232051 rad
95 percentile assoluto = 0.009374 rad
Minimo                = -0.232051 rad
Massimo               = 0.121911 rad
Deviazione standard   = 0.011428 rad

----- Statistiche a regime: campioni 150:end -----
Numero campioni      = 1852
Media                 = 0.000010 rad
Valore assoluto medio = 0.002096 rad
RMS                   = 0.004475 rad
Massimo assoluto      = 0.018743 rad
95 percentile assoluto = 0.009374 rad
Minimo                = -0.018743 rad
Massimo               = 0.018743 rad
Deviazione standard   = 0.004476 rad

```

FPGA in the Loop:

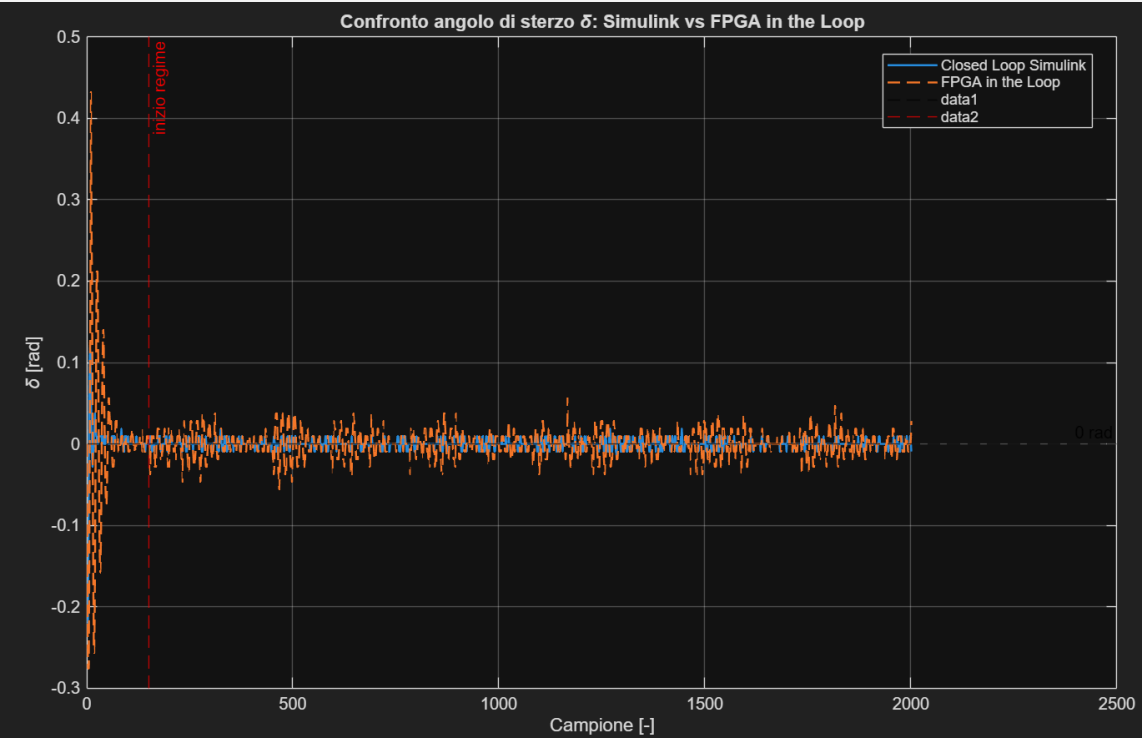
```

----- Statistiche globali: campioni 1:end -----
Numero campioni      = 2001
Media                 = 0.000006 rad
Valore assoluto medio = 0.012968 rad
RMS                   = 0.029369 rad
Massimo assoluto      = 0.431993 rad
95 percentile assoluto = 0.028102 rad
Minimo                = -0.275896 rad
Massimo               = 0.431993 rad
Deviazione standard   = 0.029376 rad

----- Statistiche a regime: campioni 150:end -----
Numero campioni      = 1852
Media                 = 0.000010 rad
Valore assoluto medio = 0.009886 rad
RMS                   = 0.013884 rad
Massimo assoluto      = 0.056068 rad
95 percentile assoluto = 0.028102 rad
Minimo                = -0.056068 rad
Massimo               = 0.056068 rad
Deviazione standard   = 0.013888 rad

```

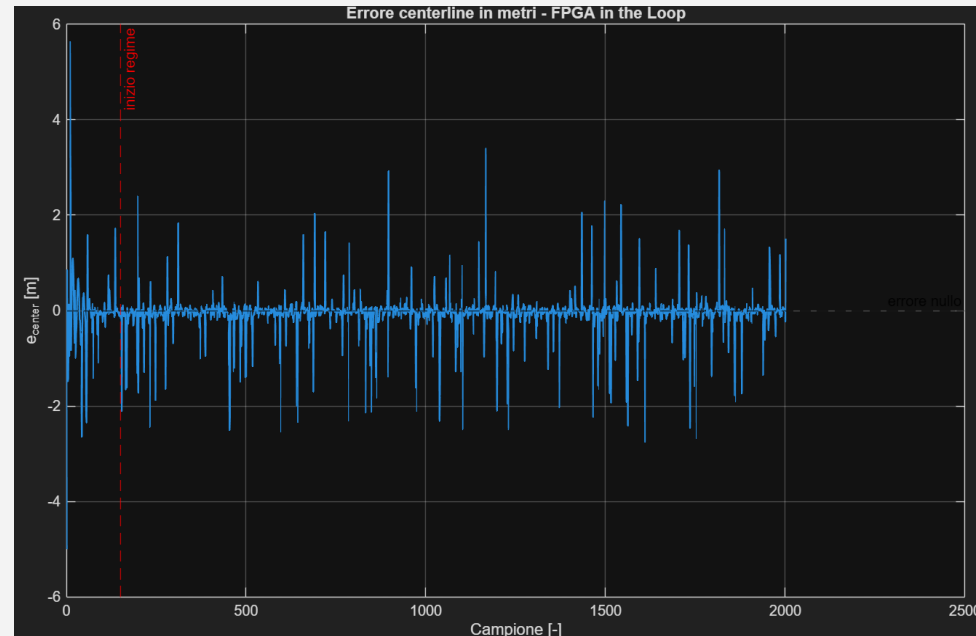
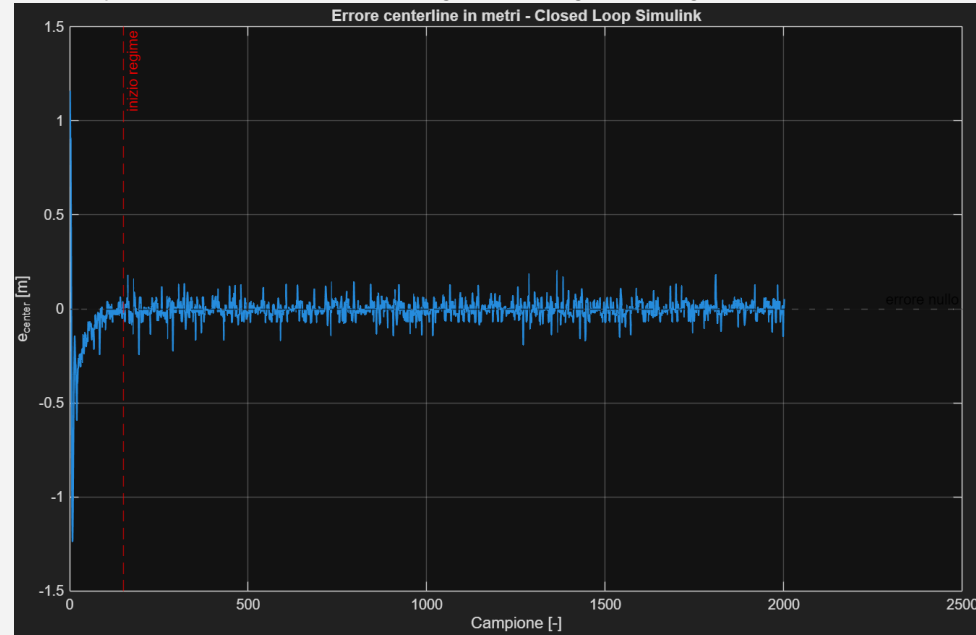
RANSAC VS QUADRATIC MODEL



Differenza delta FPGA - Simulink:

----- Statistiche globali: campioni 1:end -----	
Numero campioni	= 2001
Media	= 0.000001 rad
Valore assoluto medio	= 0.013543 rad
RMS	= 0.028225 rad
Massimo assoluto	= 0.357423 rad
95 percentile assoluto	= 0.037446 rad
Minimo	= -0.275077 rad
Massimo	= 0.357423 rad
Deviazione standard	= 0.028232 rad
----- Statistiche a regime: campioni 150:end -----	
Numero campioni	= 1852
Media	= 0.000000 rad
Valore assoluto medio	= 0.010625 rad
RMS	= 0.014642 rad
Massimo assoluto	= 0.056144 rad
95 percentile assoluto	= 0.028117 rad
Minimo	= -0.056144 rad
Massimo	= 0.056068 rad
Deviazione standard	= 0.014646 rad

RANSAC VS QUADRATIC MODEL



Closed Loop Simulink:

```

----- Statistiche globali: campioni 1:end -----
Numero campioni      = 2001
Media                 = -0.011180 m
Valore assoluto medio = 0.040288 m
RMS                   = 0.088629 m
Massimo assoluto      = 1.235752 m
95 percentile assoluto = 0.124675 m
Minimo                = -1.235752 m
Massimo               = 1.159618 m
Deviazione standard   = 0.087943 m

----- Statistiche a regime: campioni 150:end -----
Numero campioni      = 1852
Media                 = -0.002851 m
Valore assoluto medio = 0.030252 m
RMS                   = 0.042740 m
Massimo assoluto      = 0.242703 m
95 percentile assoluto = 0.078569 m
Minimo                = -0.242703 m
Massimo               = 0.201615 m
Deviazione standard   = 0.042656 m

```

FPGA in the Loop:

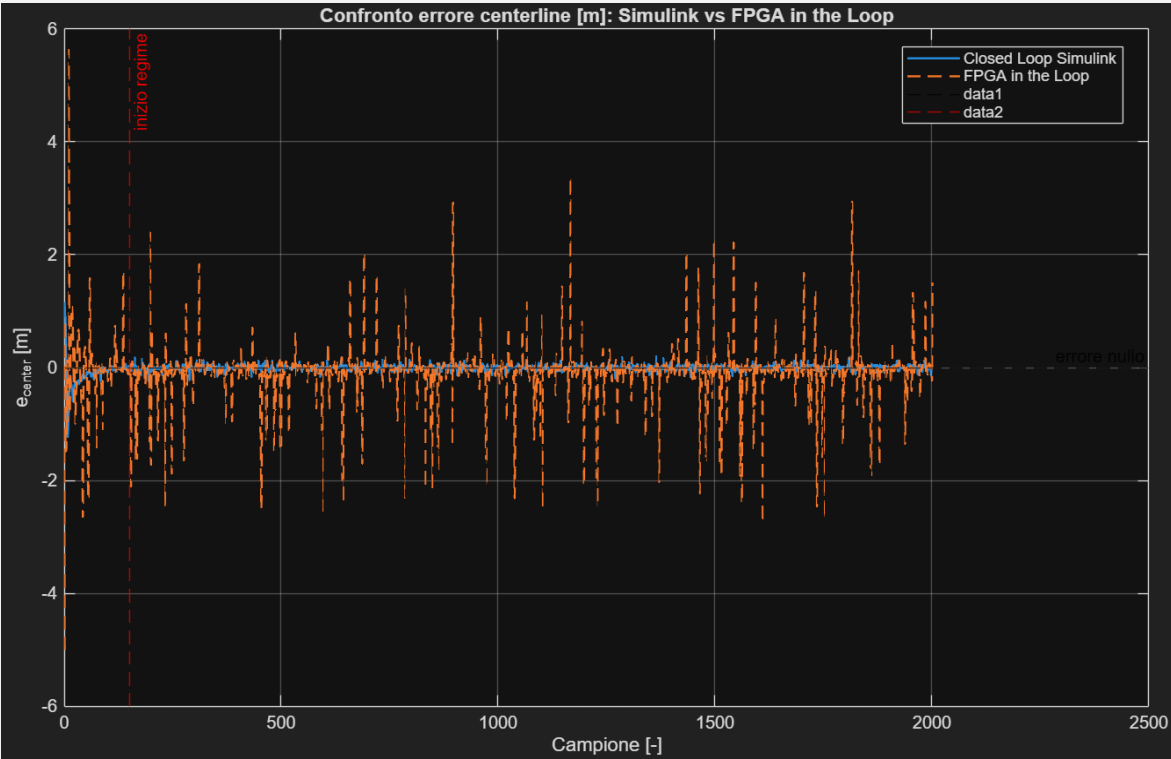
```

----- Statistiche globali: campioni 1:end -----
Numero campioni      = 2001
Media                 = -0.068160 m
Valore assoluto medio = 0.189798 m
RMS                   = 0.484717 m
Massimo assoluto      = 5.628503 m
95 percentile assoluto = 1.053752 m
Minimo                = -5.000000 m
Massimo               = 5.628503 m
Deviazione standard   = 0.480021 m

----- Statistiche a regime: campioni 150:end -----
Numero campioni      = 1852
Media                 = -0.068777 m
Valore assoluto medio = 0.169678 m
RMS                   = 0.430593 m
Massimo assoluto      = 3.391067 m
95 percentile assoluto = 0.948345 m
Minimo                = -2.748487 m
Massimo               = 3.391067 m
Deviazione standard   = 0.425180 m

```

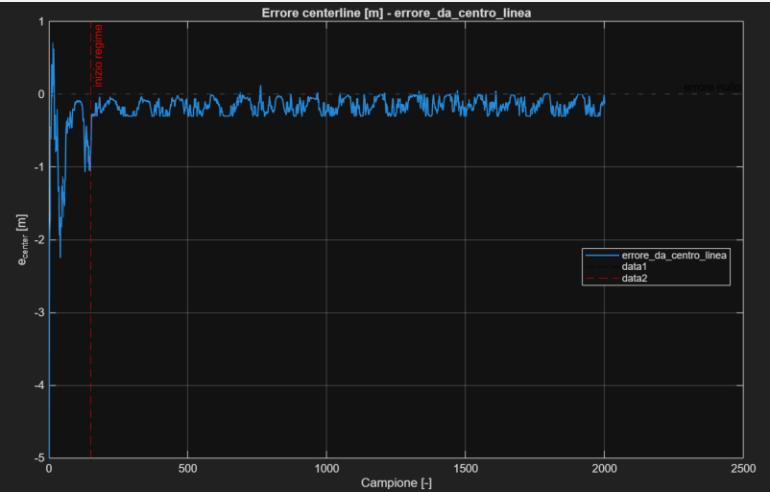
RANSAC VS QUADRATIC MODEL



Differenza FPGA - Simulink:

----- Statistiche globali: campioni 1:end -----	
Numero campioni	= 2001
Media	= -0.056981 m
Valore assoluto medio	= 0.199519 m
RMS	= 0.503801 m
Massimo assoluto	= 6.496717 m
95 percentile assoluto	= 1.130150 m
Minimo	= -6.159618 m
Massimo	= 6.496717 m
Deviazione standard	= 0.500693 m
----- Statistiche a regime: campioni 150:end -----	
Numero campioni	= 1852
Media	= -0.065926 m
Valore assoluto medio	= 0.176568 m
RMS	= 0.430654 m
Massimo assoluto	= 3.392429 m
95 percentile assoluto	= 0.962223 m
Minimo	= -2.741190 m
Massimo	= 3.392429 m
Deviazione standard	= 0.425693 m

136km/h vs 18Km/h



```
----- Statistiche a regime: campioni 150:end -----
Numero campioni      = 1852
Media                 = -0.052284 m
Valore assoluto medio = 0.052514 m
RMS                   = 0.061535 m
Massimo assoluto      = 0.255890 m
95 percentile assoluto = 0.100000 m
Minimo                = -0.255890 m
Massimo               = 0.039555 m
Deviazione standard   = 0.032456 m
```



Simulation

Curved Road



CLOSED LOOP FPGA QUADRATIC FIT



CLOSED LOOP FPGA RANSAC

FPGA PRO E CONTRO

PRO

- Elevato parallelismo
- Bassa latenza
- Comportamento temporale deterministico
- Elaborazione streaming pixel-per-pixel
- Maggiore efficienza energetica
- Integrazione diretta della telecamera
- Scalabilità dell'architettura
- Riprogrammabile
- Ridondanza: GPU + FPGA

CONTRO

- Maggiore complessità di sviluppo
- Non tutti gli algoritmi sono facilmente accelerabili
- Complessità dell'integrazione della telecamera

Sources

- [1] Slides from the course "Automation and Control in Autonomous Vehicles", held by prof. Matteo Corno, Politecnico di Milano
- [2] FPGA DataSheet: https://digilent.com/reference/_media/basys3/basys3_rm.pdf
- [3] C. Hao et al., "A Hybrid GPU + FPGA System Design for Autonomous Driving Cars," 2019 IEEE International Workshop on Signal Processing Systems (SiPS), Nanjing, China, 2019, pp. 121-126, doi: 10.1109/SiPS47522.2019.9020540.
keywords: {Field programmable gate arrays;Graphics processing units;Autonomous vehicles;Task analysis;Cameras;Road transportation;Autonomous Driving;GPU;FPGA;AI;Reliability;Hybrid System}
- [4] D. Castells-Rufas et al., "A Survey of FPGA-Based Vision Systems for Autonomous Cars," in IEEE Access, vol. 10, pp. 132525-132563, 2022, doi: 10.1109/ACCESS.2022.3230282.
keywords: {Field programmable gate arrays;Automobiles;Energy efficiency;Hardware;Autonomous automobiles;Memory management;Computer vision;Autonomous automobile;computer vision;field programmable gate arrays;reconfigurable architectures}
- [5] Chowdhury, M.R.Z.; Seum, A.; Talukder, M.R.; Amin, R.A.; Hossain, F.S.; Obermaisser, R. Towards Next-Generation FPGA-Accelerated Vision-Based Autonomous Driving: A Comprehensive Review. Signals 2025, 6, 53.
<https://doi.org/10.3390/signals6040053>
- [6] Qi Deng, Hongbin Yang, Yaqian Zhao, Ruyang Li, Gang Dong, FPGA-based heterogeneous computing framework for environment-level parallel decision-making in autonomous driving and robotic control, Expert Systems with Applications, Volume 320, 2026,132132, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2026.132132>.